

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, XIV: THE GRAPH $\mathcal{L}$ -INVARIANT AS SPECTRAL ACTION, BLOCH EFFECTIVE MASS AND ALBANESE LENGTH, AND WHAT THE CYCLIC CHERN CHARACTER SEES

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ABSTRACT. [IX] identified the graph $\mathcal{L}$ -invariant of an abelian voltage tower as $\mathcal{L}(Y_\bullet) = -\|h_\alpha\|^2$, the negative harmonic energy of the voltage cochain, and left open “a cyclic-cohomology formulation”. This paper settles the question in both directions. Negatively: for the finite-dimensional spectral triple $(C(V) \oplus C(E^+), \ell^2(V) \oplus \ell^2(E^+), D_p)$ of [II] the odd cyclic cohomology vanishes, every Fredholm index is zero, and the pairing proposed in the task specification is linear in α (it equals $4 \sum_e \alpha(e)$); a non-integer, quadratic quantity is not an index and does not pair with HC^1 . Positively: $\mathcal{L}(Y_\bullet)$ is the second variation of the spectral action of D_p along the flat $U(1)$ -connection φ_α , in the infrared,

$$\mathcal{L}(Y_\bullet) = \frac{n}{2} \lim_{t \rightarrow \infty} \frac{1}{t} \partial_\varphi^2 \operatorname{Tr} e^{-t\Delta_\varphi} \Big|_{\varphi=0},$$

equivalently minus $n/2$ times the effective mass $\lambda_0''(0)$ of the bottom Bloch band of the $\mathbb{Z}$ -cover $\tilde{Y}_\alpha$ whose finite quotients are the tower; equivalently $-m\sigma^2$, where σ^2 is the diffusion constant of the voltage along simple random walk; equivalently minus the squared length of the class $[\alpha] \in H^1(Y_0; \mathbb{Z})$ in the Jacobian torus of Y_0 , the dual of the Albanese torus of Kotani–Sunada. The ℓ -adic L -function $\Theta(T) = \det D(1+T)$ is the Floquet determinant of $\tilde{Y}_\alpha$, and its exceptional zero of order two is the parabolic bottom of the band; the Kirchhoff identity $\det G = \kappa(Y_0)$ for the cycle Gram matrix gives $\kappa(Y_0)\mathcal{L}(Y_\bullet) \in \mathbb{Z}$. All identities are exact and machine-verified on K_4 and K_5 .

STATUS OF THE RESULTS

Every numbered statement is proved. The external inputs are the vanishing of odd cyclic cohomology of finite-dimensional commutative semisimple algebras [6, Ch. 2], which is also verified by direct linear algebra, the Markov-chain central limit theorem for additive functionals of a finite reversible chain, and Kirchhoff’s theorem in its cycle form. The Kotani–Sunada Albanese torus [4, 5] and the spectral action [1, 3, 2] supply names and context, not steps. Section 9 says what was asked, what is proved, and what is impossible.

1. SETTING

Y_0 is a finite connected $(q+1)$ -regular graph with n vertices and $m = \frac{(q+1)n}{2}$ edges, E^+ an orientation, $r = m - n + 1$, $\partial: \mathbb{Z}^{E^+} \rightarrow \mathbb{Z}^V$, $\partial e = t(e) - o(e)$, $\Delta_0 = \partial\partial^t = (q+1)I - A$, $\kappa_0 = \kappa(Y_0)$. The Hecke–Dirac operator of [II, Def. 3.1] is

$$D_p = \begin{pmatrix} 0 & \partial \\ \partial^t & 0 \end{pmatrix} \text{ on } \ell^2(V) \oplus \ell^2(E^+), \quad D_p^2 = \Delta_0 \oplus \partial^t\partial,$$

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Paper XIV of the series; Papers I–XIII are cited as [I]–[XIII]. The verification script `cyclic_index.py` in the ancillary bundle reproduces every computational claim; its output is archived as `cyclic_index.output.txt`.

and $\mathcal{A} := C(V) \oplus C(E^+)$ acts diagonally, so $(\mathcal{A}, \ell^2(V) \oplus \ell^2(E^+), D_p)$ is an even finite-dimensional spectral triple with grading $\gamma = 1 \oplus (-1)$. A voltage $\alpha \in \mathbb{Z}^{E^+}$ has harmonic representative $h_\alpha = \Pi\alpha$, $\Pi = I - \partial^t \Delta_0^+ \partial$ the projection onto the cycle space $\ker D_p \cap \ell^2(E^+)$, and [IX, Def. 1.3]

$$\mathcal{L}(Y_\bullet) = -\|h_\alpha\|^2 = -(\|\alpha\|^2 - (\partial\alpha)^t \Delta_0^+ (\partial\alpha)) = \frac{\Theta''(0)}{2\kappa_0}, \quad \Theta(T) = \det D(1+T),$$

with $D(t) = (q+1)I - A(t)$, $A(t)_{uv} = \sum_{e: u \rightarrow v} t^{\alpha(e)}$, the voltage pencil. On K_4 the values are $-\frac{1}{2}, -1, -\frac{3}{2}$ for $(1, 0, 0), (1, 1, 1), (1, 2, 0)$ [IX, Table 1]; on K_5 ($q = 3, \kappa_0 = 125$) they are $-\frac{3}{5}$ for a single edge and $-\frac{7}{5}$ for a triangle (Table 1).

2. WHAT CYCLIC COHOMOLOGY SEES, AND WHAT IT CANNOT

Proposition 2.1 (Three impossibilities). (i) $HC^1(\mathcal{A}) = 0$; more precisely the space $Z_\lambda^1(\mathcal{A})$ of cyclic 1-cocycles is zero, so there is no class $\text{Ch}^1(D_p) \in HC^1(\mathcal{A})$, and $HH_1(\mathcal{A}) = 0$, so there is no class $[\delta_\alpha] \in HH_1(\mathcal{A})$ either.

(ii) Every index pairing of the even Fredholm module $(\mathcal{A}, \mathcal{H}, D_p)$ with $K_0(\mathcal{A})$ or of any compression PuP with a unitary u is an integer, and every Fredholm index of an operator between finite-dimensional spaces of equal dimension is zero. $\mathcal{L}(Y_\bullet) \notin \mathbb{Z}$ in general $(-\frac{1}{2}, -\frac{3}{2}, -\frac{3}{5}, -\frac{7}{5})$, so it is not an index.

(iii) With α acting by multiplication on $\ell^2(E^+)$,

$$\lim_{t \rightarrow 0^+} \text{Tr}_s([D_p, \alpha] D_p e^{-tD_p^2}) = \text{Tr}_s([D_p, \alpha] D_p) = 2 \text{Tr}(\alpha \partial^t \partial) = 4 \sum_{e \in E^+} \alpha(e),$$

linear in α : the formula proposed in the specification does not compute a quadratic invariant.

Proof. (i) $\mathcal{A} \cong \mathbb{C}^N$, $N = n + m$, with minimal idempotents e_i . A cyclic 1-cocycle τ satisfies $\tau(a, b) = -\tau(b, a)$ and $\tau(ab, c) - \tau(a, bc) + \tau(ca, b) = 0$. Taking $(a, b, c) = (e_i, e_i, e_k)$ with $k \neq i$ gives $\tau(e_i, e_k) = 0$, and cyclicity gives $\tau(e_i, e_i) = 0$; so $\tau = 0$. Since $\mathcal{A}$ is commutative, $b: C^0 \rightarrow C^1$ is zero and $HC^1 = Z_\lambda^1 = 0$. This is the case $n = 0$ of $HC^{2n+1}(\mathbb{C}^N) = 0$ [6, 2.1.12]; the linear system is solved by machine for $N = 10, 15$ (Table 1, line (C)). $HH_1(\mathbb{C}^N) = \Omega_{\mathbb{C}^N/\mathbb{C}}^1 = 0$ because

$\mathbb{C}^N$ is separable. (ii) is linear algebra. (iii) $[D_p, \alpha] D_p = \begin{pmatrix} \partial\alpha\partial^t & 0 \\ 0 & -\alpha\partial^t\partial \end{pmatrix}$, whose supertrace is $\text{Tr}(\partial\alpha\partial^t) + \text{Tr}(\alpha\partial^t\partial) = 2 \text{Tr}(\alpha\partial^t\partial)$, and $(\partial^t\partial)_{ee} = 2$. Continuity in t gives the limit. Verified on all six assignments: 4, 12, 12, 4, 12. $\square$

Remark 2.2 (What the even Chern character does see). $HC^{\text{even}}(\mathcal{A}) = HC^0(\mathcal{A})$ is the space of traces, $\mathbb{C}^N$, and the Chern character of the even Fredholm module $(\mathcal{A}, \mathcal{H}, D_p)$ pairs with $K_0(\mathcal{A}) = \mathbb{Z}^N$ to give the index of D_p localized at each idempotent — the content of [II, Thm. 3.3]: $\ker D_p \cong \mathbb{Z} \oplus \mathbb{Z}^r$. This is topology $(b_0$ and $b_1)$ and does not depend on α . The $\mathcal{L}$ -invariant depends on the metric (the unit-weight inner product on edges) and on α quadratically; it lives in the spectral, not the cohomological, side of the triple. Section 5 says where.

3. THE HEAT-KERNEL FORMULAS

Theorem 3.1 (Infrared, not ultraviolet). For every voltage α ,

$$\begin{aligned} \|h_\alpha\|^2 &= \lim_{t \rightarrow \infty} \langle \alpha, e^{-tD_p^2} \alpha \rangle = \|\alpha\|^2 - \int_0^\infty \langle \partial\alpha, e^{-s\Delta_0} \partial\alpha \rangle ds, \\ \langle \alpha, e^{-tD_p^2} \alpha \rangle &= \|h_\alpha\|^2 + \sum_{\lambda > 0} e^{-t\lambda} \|P_\lambda \alpha\|^2, \end{aligned}$$

the sum over the nonzero eigenvalues λ of $\partial^t \partial$ (equivalently of Δ_0) with spectral projections P_λ . The ultraviolet limit $t \rightarrow 0^+$ gives $\|\alpha\|^2$, not $\|h_\alpha\|^2$.

Proof. Π is the projection onto $\ker \partial^t \partial$, so $e^{-t\partial^t \partial} \rightarrow \Pi$ strongly and the spectral expansion follows. For the integral, $\Delta_0^+ = \int_0^\infty (e^{-s\Delta_0} - P_{\ker \Delta_0}) ds$ and $\partial\alpha \perp \mathbf{1} = \ker \Delta_0$. On K_4 : $\langle \alpha, e^{-tD_p^2} \alpha \rangle = \frac{1}{2} + \frac{1}{2}e^{-4t}$, $1 + 2e^{-4t}$, $\frac{3}{2} + \frac{7}{2}e^{-4t}$; on K_5 : $\frac{3}{5} + \frac{2}{5}e^{-5t}$, $\frac{7}{5} + \frac{8}{5}e^{-5t}$ (Table 1, line (H), symbolic in t). $\square$

4. THE $\mathbb{Z}$ -COVER, ITS BLOCH BANDS, AND THE EFFECTIVE MASS

Definition 4.1 (The $\mathbb{Z}$ -cover of the tower). $\tilde{Y}_\alpha$ is the derived graph of (Y_0, α) with voltage group $\mathbb{Z}$: vertices $V \times \mathbb{Z}$, an edge $(o(e), j) \rightarrow (t(e), j + \alpha(e))$ for each $e \in E^+$. The level- k graph Y_k of the tower [IV, §1] is $\tilde{Y}_\alpha / \ell^k \mathbb{Z}$. Its Laplacian $\tilde{\Delta}$ is $\mathbb{Z}$ -periodic, and the Floquet transform gives $\ell^2(\tilde{Y}_\alpha) \cong \int_{[0, 2\pi)}^\oplus \mathbb{C}^n \frac{d\varphi}{2\pi}$, $\tilde{\Delta} \cong \int^\oplus \Delta_\varphi$, $\Delta_\varphi := D(e^{i\varphi})$, the Hermitian twisted Laplacian. Its eigenvalues $\lambda_0(\varphi) \leq \dots \leq \lambda_{n-1}(\varphi)$ are the *Bloch bands*. The block-character decomposition [IV, Lem. 1.1] says

$$\text{spec } \Delta(Y_k) = \bigcup_{\varphi \in 2\pi\ell^{-k}\mathbb{Z}/2\pi\mathbb{Z}} \text{spec } \Delta_\varphi, \quad \Theta(e^{i\varphi} - 1) = \det \Delta_\varphi,$$

so the tower spectra sample the bands at the ℓ^k -th roots of unity and the ℓ -adic L -function is the Floquet determinant of the $\mathbb{Z}$ -cover.

Theorem 4.2 (Effective mass). λ_0 is even in φ , simple near $\varphi = 0$, and

$$\lambda_0(\varphi) = \frac{\|h_\alpha\|^2}{n} \varphi^2 + O(\varphi^4), \quad \text{i.e.} \quad \mathcal{L}(Y_\bullet) = -\frac{n}{2} \lambda_0''(0).$$

The exceptional zero of order two of Θ [IX, Thm. 1.2] is the parabolic bottom of the band: $\det \Delta_\varphi = \lambda_0(\varphi) \prod_{j \geq 1} \lambda_j(\varphi) = n\kappa_0 \cdot \frac{\|h_\alpha\|^2}{n} \varphi^2 + O(\varphi^3)$, and $\varphi^2 = -T^2 + O(T^3)$ for $T = e^{i\varphi} - 1$ gives $a_2 = -\kappa_0 \|h_\alpha\|^2$ again.

Proof. $D(t^{-1}) = D(t)^t$ [IX, Lem. 1.1] gives $\text{spec } \Delta_{-\varphi} = \text{spec } \Delta_\varphi$, so λ_0 is even and the odd Taylor terms vanish. At $\varphi = 0$ the kernel of Δ_0 is $\mathbb{C}\mathbf{1}$, simple, with gap $\lambda_1(0) > 0$; so λ_0 is analytic near 0 and Rayleigh–Schrödinger applies. Write $\Delta_\varphi = \Delta_0 + \varphi D_1 + \varphi^2 D_2 + O(\varphi^3)$ with $(D_1)_{uv} = -\sum_{e: u \rightarrow v} i\alpha(e)$, $(D_2)_{uv} = \sum_{e: u \rightarrow v} \alpha(e)^2/2$ (both sums over both orientations). Then $(D_1 \mathbf{1})_u = i(\partial\alpha)_u$, $\langle \mathbf{1}, D_1 \mathbf{1} \rangle = 0$, $\langle \mathbf{1}, D_2 \mathbf{1} \rangle = \|\alpha\|^2$, and

$$\lambda_0(\varphi) = \frac{\varphi^2}{n} \left(\langle \mathbf{1}, D_2 \mathbf{1} \rangle - \langle D_1 \mathbf{1}, \Delta_0^+ D_1 \mathbf{1} \rangle \right) + O(\varphi^4) = \frac{\varphi^2}{n} (\|\alpha\|^2 - (\partial\alpha)^t \Delta_0^+ \partial\alpha) + O(\varphi^4).$$

This is the Schur-complement branch $\lambda(1+s) = -\frac{\|h_\alpha\|^2}{n} s^2$ of [IX, proof of Thm. 3.4] with $s = e^{i\varphi} - 1$. The coefficient is computed exactly and checked to 10^{-12} numerically on all six assignments $(\frac{1}{8}, \frac{1}{4}, \frac{3}{8}, \frac{3}{25}, \frac{7}{25})$; Table 1, line (B)). $\square$

5. THE SPECTRAL ACTION

$\Delta_\varphi = \partial_\varphi \partial_\varphi^*$ with $\partial_\varphi = \partial U_\varphi$, $U_\varphi = \text{diag}(e^{i\varphi\alpha(e)})$ on $\ell^2(E^+)$, is the Hodge Laplacian of the flat $U(1)$ -connection with holonomy $e^{i\varphi\alpha(C)}$ around each cycle C ; the twisted Dirac operator is $D_p^\varphi = \begin{pmatrix} 0 & \partial U_\varphi \\ U_\varphi^* \partial^t & 0 \end{pmatrix}$, $(D_p^\varphi)^2 = \Delta_\varphi \oplus U_\varphi^* \partial^t \partial U_\varphi$, and the *spectral action* [1] of the fluctuated triple is $\text{Tr } f((D_p^\varphi)^2)$.

Theorem 5.1 ($\mathcal{L}$ as second variation of the spectral action). For the heat cutoff $f(x) = e^{-tx}$,

$$\mathcal{L}(Y_\bullet) = \frac{n}{2} \lim_{t \rightarrow \infty} \frac{1}{t} \partial_\varphi^2 \text{Tr } e^{-t\Delta_\varphi} \Big|_{\varphi=0} = \frac{n}{4} \lim_{t \rightarrow \infty} \frac{1}{t} \partial_\varphi^2 \text{Tr } e^{-t(D_p^\varphi)^2} \Big|_{\varphi=0},$$

with error $O(t e^{-t\lambda_1(0)})$. For an arbitrary smooth cutoff f ,

$$\partial_\varphi^2 \text{Tr } f(\Delta_\varphi) \Big|_0 = f'(0) \lambda_0''(0) + \sum_{j \geq 1} \partial_\varphi^2 f(\lambda_j(\varphi)) \Big|_0,$$

and $\mathcal{L}(Y_\bullet) = -\frac{n}{2}$ times the coefficient of $f'(0)$ in the ground-band term.

Proof. $\text{Tr } e^{-t\Delta_\varphi} = e^{-t\lambda_0(\varphi)} + \sum_{j \geq 1} e^{-t\lambda_j(\varphi)}$, the first term analytic near 0 by Theorem 4.2, the remainder smooth in φ (a symmetric function of the other eigenvalues of an analytic Hermitian family) with φ -derivatives bounded by $Ct^2 e^{-t\lambda_1(0)}$ near 0. Since $\lambda'_0(0) = 0$, $\partial_\varphi^2 e^{-t\lambda_0}|_0 = -t\lambda''_0(0) = -2t\|h_\alpha\|^2/n$. For the Dirac version, $(D_p^\varphi)^2 = \partial_\varphi \partial_\varphi^* \oplus \partial_\varphi^* \partial_\varphi$, and the two blocks have the same nonzero spectrum, so $\text{Tr } e^{-t(D_p^\varphi)^2} = 2 \text{Tr } e^{-t\Delta_\varphi} + (m-n)$. The general- f statement is the chain rule. Numerically on all six assignments the $t = 10$ value agrees with $\mathcal{L}$ to 10^{-14} (Table 1, line (S)). $\square$

Remark 5.2 (Reading). The Chern character of [3, 2] is the ultraviolet ($t \rightarrow 0$), cohomological, integer-valued side of the spectral triple; Proposition 2.1 shows that side is blind to α . The $\mathcal{L}$ -invariant is on the infrared ($t \rightarrow \infty$), spectral, rational side: the curvature of the ground state energy along the moduli of flat connections. On a Riemannian manifold the analogous quantity is the Hessian of the bottom of the spectrum of the twisted Laplacian at the trivial character, which is the Albanese metric [4]; the graph statement is Theorem 7.1 below. The Greenberg–Stevens shape of [IX, Rem. 3.5] — $\mathcal{L}$ as the ratio of a second character derivative to a first temperature derivative — is here the statement that the band is parabolic while the Perron eigenvalue moves linearly with β .

6. DIFFUSION

Let (X_i) be simple random walk on Y_0 started from the uniform distribution, and let

$$S_N = \sum_{i < N} \alpha(X_i, X_{i+1}), \quad \alpha(y, x) = -\alpha(x, y),$$

be the accumulated voltage; on the $\mathbb{Z}$ -cover, S_N is the fibre coordinate of the lifted walk.

Theorem 6.1 ($\mathcal{L}$ as diffusion constant). $\mathbb{E}S_N = 0$, $\text{Var}(S_N) = N\sigma^2 + O(1)$ and $S_N/\sqrt{N} \Rightarrow \mathcal{N}(0, \sigma^2)$ with

$$\sigma^2 = \frac{\|h_\alpha\|^2}{m} = -\frac{\mathcal{L}(Y_\bullet)}{m}.$$

Proof. The chain is reversible with uniform stationary law π and transition matrix $P = A/(q+1)$. Let $F(x) = \sum_y P(x, y)\alpha(x, y) = -(\partial\alpha)(x)/(q+1)$ be the drift; $\pi(F) = 0$. The Poisson equation $(I - P)g = F$, i.e. $\Delta_0 g = -\partial\alpha$, has the solution $g = -\Delta_0^+ \partial\alpha$, and $M_i = \alpha(X_i, X_{i+1}) + g(X_{i+1}) - g(X_i)$ are stationary martingale increments with $S_N = \sum M_i + g(X_0) - g(X_N)$. The martingale CLT gives the limit law with $\sigma^2 = \mathbb{E}_\pi M_0^2 = \sum_{x,y} \pi(x)P(x, y)(\alpha + \partial^t g)(x, y)^2$. Now $\alpha + \partial^t g = \alpha - \partial^t \Delta_0^+ \partial\alpha = h_\alpha$, and summing over both orientations of each edge, $\sigma^2 = \frac{2}{n(q+1)}\|h_\alpha\|^2 = \|h_\alpha\|^2/m$. The $O(1)$ term is $2\mathbb{E}[g(X_0)^2] - 2\mathbb{E}[g(X_0)g(X_N)] + \dots$, convergent geometrically. The Poisson-equation value of σ^2 is computed exactly, and $\text{Var}(S_N)$ is computed exactly for $N \leq 24$ from the moment-generating matrix over $\mathbb{Q}[s]/(s^3)$: on K_4 , $\sigma^2 = \frac{1}{12}, \frac{1}{6}, \frac{1}{4}$ and $\text{Var}(S_N) - N\sigma^2 \rightarrow \frac{1}{16}, \frac{1}{4}, \frac{7}{16}$; on K_5 , $\sigma^2 = \frac{3}{50}, \frac{7}{50}$ (Table 1, line (V)). $\square$

This is the one-dimensional case of the Kotani–Sunada central limit theorem for periodic random walks on crystal lattices [4], where the covariance is the Albanese metric.

7. ALBANESE LENGTH AND INTEGRALITY

Let $c_1, \dots, c_r$ be a $\mathbb{Z}$ -basis of the cycle lattice $H_1(Y_0; \mathbb{Z}) = \ker \partial \cap \mathbb{Z}^{E^+}$, $C = [c_1 \cdots c_r]$, $G = C^t C$ its Gram matrix, and $a_c = C^t \alpha$ the vector of periods $\langle \alpha, c_j \rangle$, which depends only on the class $[\alpha] \in H^1(Y_0; \mathbb{Z}) = \text{Hom}(H_1(Y_0; \mathbb{Z}), \mathbb{Z})$.

Theorem 7.1 ($\mathcal{L}$ as Albanese length).

$$\|h_\alpha\|^2 = a_c^t G^{-1} a_c = \|[\alpha]\|_{H^1}^2,$$

the squared norm of $[\alpha]$ in $H^1(Y_0; \mathbb{R})$ for the inner product dual to the cycle inner product on $H_1(Y_0; \mathbb{R}) \subset \mathbb{R}^{E^+}$ — the flat metric of the Jacobian torus $H^1(Y_0; \mathbb{R})/H^1(Y_0; \mathbb{Z})$, dual to the Albanese torus $\text{Alb}(Y_0) = H_1(Y_0; \mathbb{R})/H_1(Y_0; \mathbb{Z})$ of [5] in the unit-weight normalization. Hence $\mathcal{L}(Y_\bullet) = -\|[\alpha]\|_{H^1}^2$ depends only on $[\alpha]$, is invariant under switching $\alpha \mapsto \alpha + \partial^t \varphi$, and

$$\det G = \kappa(Y_0), \quad \kappa(Y_0) \mathcal{L}(Y_\bullet) \in \mathbb{Z},$$

which is the integrality $a_2 \in \mathbb{Z}$ of [IX, Thm. 1.2(ii)] seen from the cycle side.

Proof. h_α is the orthogonal projection of α onto the column space of C , so $h_\alpha = C(C^t C)^{-1} C^t \alpha$ and $\|h_\alpha\|^2 = \alpha^t C G^{-1} C^t \alpha$. The harmonic form h_α is the unique cycle with periods a_c , so $\|h_\alpha\|$ is the dual norm of the functional $[\alpha]$ on $(H_1(Y_0; \mathbb{R}), \langle \cdot, \cdot \rangle)$. Kirchhoff's theorem in its cycle form (the Gram determinant of a fundamental cycle basis equals the number of spanning trees) gives $\det G = \kappa_0$; a $\mathbb{Z}$ -basis change has determinant ± 1 , so $\det G = \kappa_0$ for every $\mathbb{Z}$ -basis, and $G^{-1} \in \kappa_0^{-1} M_r(\mathbb{Z})$. Verified: $\det G = 16, 125$; $\kappa_0 \mathcal{L} = -8, -16, -24, -75, -175$ (Table 1, lines (A),(T)). $\square$

Corollary 7.2 (Five faces of one number). *For an abelian voltage tower over Y_0 the following are equal: $\mathcal{L}(Y_\bullet) = \Theta''(0)/2\kappa_0$; $-\|h_\alpha\|^2$; $-\frac{n}{2} \lambda_0''(0)$; $\frac{n}{2} \lim_{t \rightarrow \infty} t^{-1} \partial_\varphi^2 \text{Tr } e^{-t\Delta_\varphi}|_0$; $-m\sigma^2$; $-\|[\alpha]\|_{H^1}^2$.*

8. THE VERIFICATION BATTERY

check	cases	result
(C) $\dim Z_\lambda^1(\mathcal{A}) = \dim HC^1(\mathcal{A}) = 0$, $\mathcal{A} = \mathbb{C}^{10}, \mathbb{C}^{15}$	K_4, K_5	exact
(N) $\text{Tr}_s([\mathbb{D}_p, \alpha] \mathbb{D}_p) = 4 \sum \alpha(e) = 4, 12, 12, 4, 12$	all six	exact
(H) $\langle \alpha, e^{-t\mathbb{D}_p^2} \alpha \rangle = \ h_\alpha\ ^2 + ce^{-t\lambda_1}$, limit and Duhamel integral $= \ h_\alpha\ ^2$	all six	exact, symbolic t
(B) $\lambda_0(\varphi) = \frac{\ h_\alpha\ ^2}{n} \varphi^2 + O(\varphi^4)$: exact coefficient $\frac{1}{8}, \frac{1}{4}, \frac{3}{8}, \frac{3}{25}, \frac{7}{25}$; Richardson numerics	all six	exact + 40 digits
(S) $\frac{n}{2} t^{-1} \partial_\varphi^2 \text{Tr } e^{-t\Delta_\varphi} _0$ at $t = 5, 10, 20, 40$: $\mathcal{L}$ to $10^{-8}, 10^{-14}, 10^{-14}, 10^{-13}$	all six	numeric
(V) $\sigma^2 = \ h_\alpha\ ^2/m = \frac{1}{12}, \frac{1}{6}, \frac{1}{4}, \frac{3}{50}, \frac{7}{50}$; $\text{Var}(S_N)$ exact, $N \leq 24$	all six	exact
(A) $a_c^t G^{-1} a_c = \ h_\alpha\ ^2$; switching invariance; $\det G = \kappa_0$; $\kappa_0 \mathcal{L} \in \mathbb{Z}$	all six	exact
(T) [IX, Thm. 1.2(ii)] on K_5 : $a_2 = -75, -175 = -\kappa_0 \ h_\alpha\ ^2$	K_5	exact

TABLE 1. Verification (`cyclic_index.py`). “All six” = K_4 with $(1, 0, 0)$, $(1, 1, 1)$, $(1, 2, 0)$ [the $(1, 0, 0)$, $\ell = 3$ tower has the same voltage and $\mathcal{L}$] and K_5 with a single-edge and a triangle voltage.

9. SCOPE, RESIDUE, CORRECTIONS

Remark 9.1 (Solved, open, impossible). *Solved:* open item (a) of [IX, Rem. 5.1], “a cyclic-cohomology formulation of Theorem 3.4(iii)”, in the only form it admits: the $\mathcal{L}$ -invariant is the infrared second variation of the spectral action of the Hecke–Dirac triple along the flat-connection direction α (Theorem 5.1), the effective mass of the $\mathbb{Z}$ -cover (Theorem 4.2), its diffusion constant (Theorem 6.1), and the Albanese length of $[\alpha]$ (Theorem 7.1); the exceptional zero is the parabolic band bottom, and $\kappa_0 \mathcal{L} \in \mathbb{Z}$ is Kirchhoff. *Open:* (a) whether a nonzero odd cyclic class on a genuinely non-commutative algebra of the tower, such as the crossed product $C_0(\tilde{Y}_\alpha) \rtimes \mathbb{Z}$ or the Cuntz–Krieger algebra of [VIII], pairs to $\|h_\alpha\|^2$ or to its integer multiple $\kappa_0 \|h_\alpha\|^2$; the integrality of Theorem 7.1 makes the latter the only candidate for an index. (b) The rank-two analogue for the $\tilde{A}_2$ towers of [X]–[XII], where $b_1 = 0$ [XII] and there is no α : the Bloch picture survives (the Hecke

pencil of $[X]$ is a Floquet determinant over a 2-torus when a $\mathbb{Z}^2$ -cover exists) but the $\mathcal{L}$ -invariant does not. *Impossible as stated:* Proposition 2.1, all three items.

Remark 9.2 (Corrections to the task specification). (1) There is no “ $\text{Ch}^1(D_p) \in HC^1(\mathcal{A})$ ” and no “ $[\delta_\alpha] \in H_1(\mathcal{A})$ ”: both groups vanish for $\mathcal{A} = C(V) \oplus C(E^+)$, and every derivation of $\mathcal{A}$ is zero. (2) The proposed pairing $\lim_{t \rightarrow 0^+} \text{Tr}_s([D_p, \alpha] D_p e^{-tD_p^2})$ equals $4 \sum_e \alpha(e)$; it is linear in α and cannot equal $-\|\Pi\alpha\|^2$. (3) The $\mathcal{L}$ -invariant is not an index: it is rational and non-integral, and finite-dimensional Fredholm indices vanish. (4) The correct limit is $t \rightarrow \infty$, not $t \rightarrow 0^+$ (Theorem 3.1); the $t \rightarrow 0^+$ side of the JLO cocycle is the topological Chern character, which does not see α . (5) The theorem that does hold is a second-variation (curvature) statement for the spectral action, Theorem 5.1, and its geometric name is the Albanese metric, not “quantum topological index”. (6) The $(1, 0, 0)$, $\ell = 3$ tower is not a separate case: $\mathcal{L}$ depends on α only, and ℓ enters through [IX, §2] alone. The battery adds K_5 ($q = 3$) as asked.

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Unrefereed preprint. Every numbered statement is proved from Papers II, IV and IX and finite linear algebra, plus the Markov-chain CLT and Kirchhoff’s theorem in cycle form. The vanishing of $HC^1(\mathbb{C}^N)$ is both quoted [6] and verified directly. Bibliographic data of [4, 5, 3, 1, 6, 2] were checked against publisher or indexing pages; [4, 5] are cited for the definition of the Albanese torus and the crystal-lattice CLT, and no statement of this paper depends on them.

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